# Counterfactual Attribution of Crypto Systemic Stress on an Auditable On-Chain Graph

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**Abstract**

Crypto systemic-risk measures usually infer contagion networks from prices. We instead use the entity-resolved on-chain ledger as an auditable candidate skeleton for a linear structural causal model of entity stress. On this skeleton we define On-Chain Causal CoVaR, an interventional measure, and a counterfactual loss attribution with an exact Shapley decomposition. Interventional effects are pure network multipliers, independent of flows and shocks. This gives flow-free bounds on the tail measure and makes zero attribution between unconnected entities a theorem. A closed-form walk representation of the linear Shapley attribution assigns negative credit to intermediaries, and an exact subset algorithm needs $O(2^p)$ rather than $O(p!\,p)$ model evaluations. Applied to Ethereum flows during the October 2025 liquidation cascade, with checks on FTX and USDC/SVB, the attribution isolates inter-exchange channels: Binance→Bybit accounts for 98% of Bybit's model-implied distress. The hourly exported-risk ranking has Spearman correlation 0.52 with $\Delta$CoVaR (six entities), and a deterministic proxy of it is stable under coefficient perturbation (0.88). Against a price-estimated graph, the observed graph is only weakly better (bootstrap interval includes zero). Direction is event-dependent: reversing edges degrades prediction in FTX and USDC/SVB, yet a symmetric graph is as good or better in several comparisons.

**Keywords:** systemic risk, cryptocurrency, structural causal model, counterfactual attribution, Shapley value, on-chain data

# 1 Introduction

On 10–11 October 2025 cryptocurrency markets experienced one of the largest deleveraging episodes on record, with on the order of nineteen billion US dollars of leveraged positions liquidated within about thirty-six hours.[1] Post-mortems (Ali 2025; Lim 2026) describe a largely mechanical cascade: a mispricing in a centralised exchange's collateral valuation triggered liquidations, which forced sales and further liquidations. Over-collateralised decentralised-finance (DeFi) lending protocols such as Aave recorded essentially no bad debt, whereas centralised-exchange (CEX) risk engines amplified the shock. The episode left a high-resolution on-chain footprint of liquidation transfers, collateral movements and exchange flows, timestamped to the block.

The standard systemic-risk tools are associational. $\Delta$CoVaR (Adrian and Brunnermeier 2016), marginal expected shortfall and SRISK (Acharya et al. 2017), and connectedness measures (Diebold and Yılmaz 2014; Billio et al. 2012) quantify co-movement in returns, and a survey of causal inference in finance finds that applications around these measures remain associational or Granger-based (Kumar et al. 2025). Where causality has been added, the contagion graph is still estimated from prices (Rigana et al. 2024, 2023; Etesami et al. 2023; Giudici and Parisi 2018). Shapley-value risk attribution (Tarashev et al. 2016) and causal Shapley methods (Heskes et al. 2020; Singal et al. 2021) come closest in spirit, but they do not take an observed transaction ledger as the causal structure of a systemic-risk measure (Section 2).

We make three contributions. First, VEIN treats the entity-resolved transaction graph as an *auditable candidate structural skeleton*: under the assumptions A1–A3 (Section 3), the in-neighbours of each entity specify its parent set in a Pearl SCM rather than a set inferred from correlations. Second, we define On-Chain Causal CoVaR (OC-CoVaR), an interventional measure, and a counterfactual attribution of realised loss. We prove that the interventional effect is a pure network multiplier, derive flow-free

[1] See CoinDesk Research, https://www.coindesk.com/research/market-spotlight-the-19-billion-liquidation-that-shook-crypto.

bounds, and give a closed-form walk decomposition of the linear Shapley attribution together with an exact subset algorithm (Propositions 2–4). Third, we implement the pipeline, including a four-tier entity-resolution layer, apply it to the October 2025 event (with FTX and USDC/SVB as additional episodes), and test the directional assumption by edge reversal and the graph source against price-estimated graphs. The output is an entity ranking and a loss decomposition that a supervisor can trace to public ledger records and an explicit model, which is the property that price-based measures lack.

## 2 Related Work and Positioning

*Associational measures.* ΔCoVaR (Adrian and Brunnermeier 2016) is the change in system value-at-risk when an institution moves from its median state to distress. The SRISK and MES family (Acharya et al. 2017) and connectedness measures (Diebold and Yılmaz 2014; Billio et al. 2012) describe spillovers in returns. They describe co-movement rather than mechanism, which is the gap the causal survey of Kumar et al. (2025) also documents.

*Estimated causal graphs.* Causal-NECO VaR (Rigana et al. 2024) learns the graph with the PC-stable algorithm (Colombo and Maathuis 2014) and computes a Gaussian VaR from the resulting structural model (Rigana et al. 2023; Spirtes et al. 2000). It operates at Pearl Levels 1–2 on foreign-exchange data and estimates its graph. The time-varying directed-information graph of Etesami et al. (2023) is the nearest crypto application. Directed information generalises Granger causality (Granger 1969) but is not a Pearl SCM with do-operator semantics, and both approaches are built from returns. CoRisk (Giudici and Parisi 2018) decomposes risk on a correlation network. Crypto composite indices (Farzulla and Maksakov 2026) and TradFi–DeFi mappings (Aufiero et al. 2025) either estimate networks from prices or describe the on-chain graph without a causal model.

*Shapley attribution and causal Shapley values.* Tarashev et al. (2016) allocate system-wide risk to institutions with the Shapley value, and Drehmann and Tarashev (2013) measure the systemic importance of interconnected banks. These allocations use exposures or portfolio losses rather than a causal model of propagation. Causal Shapley values (Heskes et al. 2020) respect a known causal graph, and Singal et al. (2021) give a recursive Shapley attribution for flows in graphical models. Both explain model outputs or flows. VEIN attributes realised systemic loss to interventions on a ledger-specified graph.

*Contagion and on-chain data.* Network models of default propagation (Eisenberg and Noe 2001; Acemoglu et al. 2015; Glasserman and Young 2016) motivate the network as the object of study. VEIN uses reduced-form linear propagation instead of a clearing mechanism. On-chain data have been used to study Tether issuance (Griffin and Shams 2020), arbitrage and capital flows (Makarov and Schoar 2020), reconstruct the Terra/Luna run (Liu et al. 2023), measure DeFi liquidations (Qin et al. 2021), and for prediction and forensics (Akcora et al. 2018; Meiklejohn et al. 2013; Weber et al. 2019). None treats the graph as a structural causal model for a systemic-risk measure.

Recent crypto work is descriptive or price-based. Sankaewtong et al. (2026) trace USDC depeg contagion through transactions without interventional semantics, Zhang et al. (2026) use correlations of total-value-locked series, and counterfactual analysis has so far reached credit-portfolio losses (Xu 2025). The claimed novelty is narrow: an observed, ledger-specified graph used as the parent set of a structural causal model, combined with a counterfactual Shapley attribution of systemic loss. To our knowledge no published work combines these two elements. Because this is a negative existence claim, it is bounded by our search coverage.

# 3 The VEIN Framework

## 3.1 Structural causal model and identification

Let $G = (V, E)$ be a directed, timestamped, dollar-weighted graph of resolved entities (exchanges, lending protocols, stablecoin issuers, custodians and an aggregate retail node) whose edges are observed on-chain flows. For entity $i$ and period $t$ we track a stress state $S_{i,t} \in \mathbb{R}$, oriented so that larger means more distressed, the flow $F_{p\to i,t} \geq 0$, and an exogenous shock $U_{i,t}$. The structural equation is

$$S_{i,t} \;=\; f_i\Big(S_{\mathrm{pa}(i),\,t-1},\; F_{\mathrm{pa}(i)\to i,\,t},\; U_{i,t}\Big), \qquad \mathrm{pa}(i) \;=\; \{\, p \in V : (p \to i) \in E \,\}. \tag{1}$$

Under assumptions A1–A3 below, the in-neighbours of $i$ specify its parent set $\mathrm{pa}(i)$ from the ledger, so only the propagation function $f_i$ requires estimation. We use a ridge-regularised linear specification,

$$S_{i,t} \;=\; b_i + \sum_{p\in\mathrm{pa}(i)} a_{ip}\, S_{p,t-1} + \sum_{p\in\mathrm{pa}(i)} c_{ip}\, \log\!\big(1 + F_{p\to i,t}\big) + U_{i,t}, \tag{2}$$

with parameters estimated on a pre-crisis window; the fitted residuals furnish the shock distribution.

Every parent enters at lag $t-1$ in the stress term, so the system is well posed even when $G$ has cycles.

**Proposition 1** (Well-posedness of the lagged system) *Given $S_{\cdot,-1}$ and a shock path, the trajectory generated by (2) is uniquely determined by forward iteration, even when $G$ contains directed cycles.*

*Proof* At step $t$ the right-hand side of (2) references only stress at $t-1$ and exogenous quantities at $t$. Induction on $t$ gives existence and uniqueness. □

Identification rests on three assumptions. *Temporal precedence* (A1): a flow $F_{i\to j,t}$ precedes any effect on $j$ at $t' > t$. Block timestamps make this observable, but they show only that a transfer could not be an effect of later stress, not that it is the operative cause. *No unobserved common cause for flow edges* (A2): given the graph up to $t$, an edge $i \to j$ reflects $i$'s direct action on $j$. This is the strongest assumption, because the ledger does not record off-chain decisions such as a CEX-internal pricing engine, an oracle update or coordinated withdrawals, any of which can drive both a flow and downstream stress. Section 5.3 bounds this threat but does not eliminate it. *Monotone propagation along flow direction* (A3): distress at $i$ affects $j$ only through realised flow paths from $i$ to $j$. A3 carries the empirical content and is tested by edge reversal.

## 3.2 Interventional and counterfactual risk measures

Over an event window $\mathcal{T}$ let $L_j = \sum_{t\in\mathcal{T}} \max(S_{j,t}, 0)$. The interventional measure applies Pearl's do-operator to (1): entity $i$ is forced to a distressed level $s^*$, its parents are severed, and the system is propagated forward. On-Chain Causal CoVaR is

$$\text{OC-CoVaR}_q\big(j \,\big|\, \text{do}(S_i = s^*)\big) \;=\; \text{VaR}_q\Big(L_j \,\Big|\, \text{do}(S_i = s^*)\Big), \tag{3}$$

estimated by Monte-Carlo propagation with residual-resampled shock paths. The marginal contribution contrasts the distressed intervention with the median state,

$$\Delta\text{OC-CoVaR}_q(j \,|\, i) \;=\; \text{OC-CoVaR}_q\big(j \,|\, \text{do}(S_i = s^*)\big) - \text{OC-CoVaR}_q\big(j \,|\, \text{do}(S_i = 0)\big), \tag{4}$$

with $s^*$ the 95th percentile of $i$'s stress. Entities are ranked by exported tail risk $\mathcal{E}(i) = \sum_{j\neq i} \max(\Delta\text{OC-CoVaR}_q(j \,|\, i), 0)$.

**Algorithm 1** VEIN end-to-end pipeline

1: $G \leftarrow$ ResolveEntities(transfers, labels) ▷ Tiers 0–3, Section 4
2: Fit $\hat{\theta}_i$ by ridge regression for each entity $i$ on its pre-event window
3: $\Delta$OC-CoVaR$_q(j \mid i)$ by MonteCarloPropagate under do$(S_i = s^*)$
4: $\Delta_i^{CF}(j)$ by Abduct, then Predict under do$(S_i = S_i^{\text{pre}})$
5: Rerun the three-variant comparison on $G$, $G^{\text{rev}}$ and $G^{\text{sym}}$
6: **Output:** OC-CoVaR ranking, counterfactual attributions, falsification and validity battery

The counterfactual measure sits one level higher. Following the abduction–action–prediction procedure (Pearl 2009, 2019), we replace $i$'s realised path by its pre-crisis level $S_i^{\text{pre}}$, hold the other entities' abduced shocks fixed, and predict the change in $j$'s loss. With $L_j^{\text{fact}}$ the loss in the reference (factual) world,

$$\Delta_i^{CF}(j) \;=\; L_j^{\text{fact}} - L_j^{\,\text{do}(S_i = S_i^{\text{pre}})}, \qquad \text{share}_i(j) \;=\; \frac{\Delta_i^{CF}(j)}{L_j^{\text{fact}}}. \tag{5}$$

The factual world is model-implied: entity $i$ follows its realised path and the other entities follow the fitted equations with the abduced shocks. In the reported analysis the event-window shocks are zero (the legacy abduction, Appendix A), so $L^{\text{fact}}$ is model-implied rather than an exactly reproduced loss. By Proposition 3, this choice affects the attribution only through the hinge in $L$. The attribution is model-relative: it holds the other entities' abduced shocks fixed rather than solving the cross-world equilibrium.

### 3.3 Analytic properties and exact attribution

Let $n = |V|$ and let $A \in \mathbb{R}^{n \times n}$ satisfy $A_{jp} = a_{jp}$ for $p \in \text{pa}(j)$ and $A_{jp} = 0$ otherwise. Over the window $t = 0, \ldots, T-1$, with a fixed initial state, (2) is the linear system

$$S_t = A\, S_{t-1} + \kappa_t, \tag{6}$$

where $\kappa_{j,t} = b_j + \sum_p c_{jp} \log(1 + F_{p \to j,t}) + U_{j,t}$ collects the intercepts, flows and shocks. For an entity $i$ define the *impulse* $\tau_0^{(i)} = e_i$ and, for $t \geq 1$, $\tau_t^{(i)} = A\tau_{t-1}^{(i)}$ with its $i$-th coordinate reset to one. Write $\tau_{ji}(t)$ for the $j$-th coordinate. The impulse is the per-unit response of $j$ at $t$ to holding $i$ at a unit distressed level from the start of the window. Expanding the recursion shows that $\tau_{ji}(t)$ is a sum over walks from $i$ to $j$ of length at most $t$, each walk weighted by the product of its coefficients and not revisiting $i$. In particular $\tau_{ji}(t) \neq 0$ requires a directed path from $i$ to $j$ of length at most $t$.

**Proposition 2** (Flow-free bounds on the intervention) *Fix $i$, $T$ and $s > 0$. Let $S^{(s)}$ be the window path under $\mathrm{do}(S_{i,t} = s,\ t < T)$ and $S^{(0)}$ the path under $\mathrm{do}(S_{i,t} = 0,\ t < T)$, both with the same initial state and the same inputs $\kappa$. Then:*

*(a) $S_{j,t}^{(s)} - S_{j,t}^{(0)} = s\,\tau_{ji}(t)$ for all $j$ and $t < T$, for every realisation of $\kappa$;*

*(b) for every $q \in (0,1)$, every law of $\kappa$, and every flow path,*

$$\sum_{t<T} \min\big(s\tau_{ji}(t), 0\big) \ \leq\ \Delta OC\text{-}CoVaR_q(j \,|\, i) \ \leq\ \sum_{t<T} \max\big(s\tau_{ji}(t), 0\big), \tag{7}$$

*where $\Delta OC\text{-}CoVaR_q(j \,|\, i)$ is defined as in (4) with $s$ in place of $s^*$, and the bounds also hold for the Monte-Carlo estimator when both interventions use the same shock draws.*

**Corollary 1** (Per-period multiplier) *For $t < T$, $\mathrm{VaR}_q(S_{j,t}^{(s)}) - \mathrm{VaR}_q(S_{j,t}^{(0)}) = s\,\tau_{ji}(t)$ for every $q$ and every law of $\kappa$ (immediate from Proposition 2(a)). The per-period effect is a deterministic network multiplier.*

**Corollary 2** (Zero by construction) *If there is no directed path from $i$ to $j$ of length at most $T-1$, then $\tau_{ji}(t) = 0$ for all $t < T$, so $\Delta OC\text{-}CoVaR_q(j \,|\, i) = 0$ for every $q$ and every law of $\kappa$. The same holds for the counterfactual attribution, $\Delta_i^{CF}(j) = 0$.*

**Corollary 3** (Exported-risk bound and long-run multiplier) *Let $\Lambda_i(T) = \sum_{j \neq i} \sum_{t<T} \max(\tau_{ji}(t), 0)$. Then (i) $\mathcal{E}(i) \leq s\,\Lambda_i(T)$, and $\Lambda_i(T) = 0$ implies $\mathcal{E}(i) = 0$;*

*(ii) if* $\rho(A_{-i,-i}) < 1$, *then* $\tau_{\cdot i}(t) \to m^{(i)} := (I - A_{-i,-i})^{-1} A_{-i,i}$ *as* $t \to \infty$, *so* $\Lambda_i(T)/T \to \sum_{j \neq i} \max(m_j^{(i)}, 0)$.

Proposition 2 implies that the interventional effect is a pure network multiplier: it depends on the graph and the coefficients, not on flows, intercepts or the shock law. Proofs are in Appendix C.

**Proposition 3** (Abduction invariance of the stress deviation) *Let* $S^{\mathrm{f}}$ *be a reference path that satisfies* (6) *for entities outside a set* $K$, *generated with inputs* $\kappa$. *For* $k \in K$ *let* $p_{k,t}$ *be prescribed levels, and define* $g_{k,t} = S^{\mathrm{f}}_{k,t} - p_{k,t}$. *Let* $S^K$ *be the path under* $\mathrm{do}(S_{k,t} = p_{k,t},\, k \in K)$ *with the same initial state and inputs. Then* $S^K = S^{\mathrm{f}} - r(K)$, *where* $r$ *is the linear propagation*

$$r_{k,t} = g_{k,t} \ (k \in K), \qquad r_{j,t} = \sum_p a_{jp}\, r_{p,t-1} \ (j \notin K), \qquad r_{\cdot,-1} = 0. \tag{8}$$

*Consequently, for the counterfactual* $p_{i,t} \equiv S^{\mathrm{pre}}_i$,

$$\Delta_i^{CF}(j) = \sum_{t<T} \Big[ \max\left(S^{\mathrm{f}}_{j,t}, 0\right) - \max\left(S^{\mathrm{f}}_{j,t} - r_{j,t}(\{i\}), 0\right) \Big]. \tag{9}$$

*The stress deviation* $r$ *does not depend on the flows, intercepts or shock values except through the reference levels at the intervened entities. The only nonlinearity in the attribution is the hinge* $\max(\cdot, 0)$.

Proposition 3 separates the attribution into a linear propagation and a hinge.

**Proposition 4** (Walk representation and exact linear Shapley attribution) *Let* $P \subseteq V \setminus \{j\}$ *be the set of candidate sources and* $g_{k,t}$ *as in Proposition 3, with* $K$ *replaced by subsets of* $P$. *Consider the two games on* $P$:

$$v(K) = \sum_{t<T} r_{j,t}(K), \qquad v_H(K) = \sum_{t<T} \Big[ \max(S^{\mathrm{f}}_{j,t}, 0) - \max\left(S^{\mathrm{f}}_{j,t} - r_{j,t}(K), 0\right) \Big],$$

*where* $r(K)$ *is defined by* (8) *with* $K$ *as the intervened set. Let a* walk *be a sequence* $(x_0, t_0), (x_1, t_0+1), \ldots, (x_\ell, t_0+\ell)$ *with* $x_\ell = j$, $t_0 + \ell < T$, *and each step along an edge of* $G$;

*let* $\omega(w) = g_{x_0,t_0} \prod_s a_{x_{s+1}x_s}$, *let* $I(w) = \{x_1, \ldots, x_{\ell-1}\} \cap P$ *be the distinct candidate sources among the intermediate nodes, and* $m(w) = |I(w)|$. *Let* $\mathcal{W}$ *be the set of walks with* $x_0 \in P$ *that do not revisit* $x_0$. *Then:*

*(a)* $v(K) = \sum_{w \in \mathcal{W}} \omega(w)\, \mathbf{1}[x_0 \in K,\ I(w) \cap K = \emptyset]$;

*(b) the Shapley value of* $v$ *is, for* $k \in P$,

$$\phi_k(v) = \sum_{w \in \mathcal{W}:\, x_0 = k} \frac{\omega(w)}{1 + m(w)} \;-\; \sum_{w \in \mathcal{W}:\, k \in I(w)} \frac{\omega(w)}{m(w)\,(m(w)+1)}; \tag{10}$$

*(c)* $\sum_{k \in P} \phi_k(v) = v(P) = \sum_{w \in \mathcal{W}:\, I(w) = \emptyset} \omega(w)$;

*(d) if* $S^{\mathrm{f}}_{j,t} \geq 0$ *and* $S^{\mathrm{f}}_{j,t} - r_{j,t}(K) \geq 0$ *for all* $K \subseteq P$ *and* $t < T$, *then* $v_H = v$, *so* (10) *gives the exact Shapley values of the hinge game.*

The games are defined through $r(K)$. By Proposition 3, $v(K)$ is the change in the cumulative stress of $j$ under do on $K$ when the reference path satisfies the structural equations at every entity outside $K$. The second term of (10) is the negative credit that an intermediate candidate source receives on walks that pass through it. A walk that reaches $j$ through a second candidate source contributes positively to the originating source and negatively to the intermediary. The games here let unselected entities propagate, as in (8). The Bybit Shapley values in Section 5.2 use a different game, in which every non-candidate entity is held at its pre-crisis level, so Proposition 4 does not describe them. The walk representation is verified numerically on synthetic paths only (Section 5.5). Proposition 4(d) gives the closed form of the hinge game whenever the hinge is inactive.

Shapley's definition averages marginal contributions over the $p!$ orderings of the players (Shapley 1953). Algorithm 2 evaluates the game once on each of the $2^p$ subsets and weights the marginals by the number of orderings that place each coalition first. It is exact and uses no sampling. The number of oracle calls falls from $p!\,p$ for permutation enumeration to $2^p$. Each call costs $O(T(|V|+|E|))$ by forward propagation. For $p = 13$

**Algorithm 2** Exact subset Shapley attribution

**Input:** candidate sources $P$ with $p = |P|$; value oracle $v : 2^P \to \mathbb{R}$ (a linear or hinge window propagation)
**for** each $K \subseteq P$ **do**
    $v(K) \leftarrow$ oracle evaluation (one forward propagation of the window)
**end for**
**for** each $k \in P$ **do**
    $\phi_k \leftarrow \sum_{K \subseteq P \setminus \{k\}} \frac{|K|!\,(p-|K|-1)!}{p!} \big(v(K \cup \{k\}) - v(K)\big)$
**end for**
**Output:** $\{\phi_k\}_{k \in P}$ with $\sum_k \phi_k = v(P) - v(\emptyset)$

the saving is 8,192 calls against 80,951,270,400 permutation evaluations, which is the comparison used in Section 5.5.

### 3.4 Validation design, sensitivity and baselines

To test A3 we construct three variants of $G$: the true-direction graph, a reversed graph $G^{\text{rev}}$, and a symmetric graph $G^{\text{sym}}$. Each is fitted on the estimation window and compared on one-step-ahead prediction of event-window stress by the stress-weighted root-mean-square error $\text{RMSE} = \sqrt{\sum_{i,t} w_{i,t}(\hat{S}_{i,t} - S_{i,t})^2 / \sum_{i,t} w_{i,t}}$, with $w_{i,t} = |S_{i,t}| + 0.1$. Because A2 is not directly testable, we report the robustness value of Cinelli and Hazlett (2020). For a coefficient with $t$-statistic $t$ on $d$ degrees of freedom and $f = |t|/\sqrt{d}$,

$$RV \;=\; \tfrac{1}{2}\Big(\sqrt{f^4 + 4f^2} - f^2\Big) \in [0, 1] \tag{11}$$

is the share of residual variance a confounder must explain in *both* the parent and the child to drive the coefficient to zero. Baselines are $\Delta$CoVaR (Adrian and Brunnermeier 2016), MES (Acharya et al. 2017), and, for H1, Granger and partial-correlation graphs (Granger 1969; Giudici and Parisi 2018) fed through the same VEIN machinery.

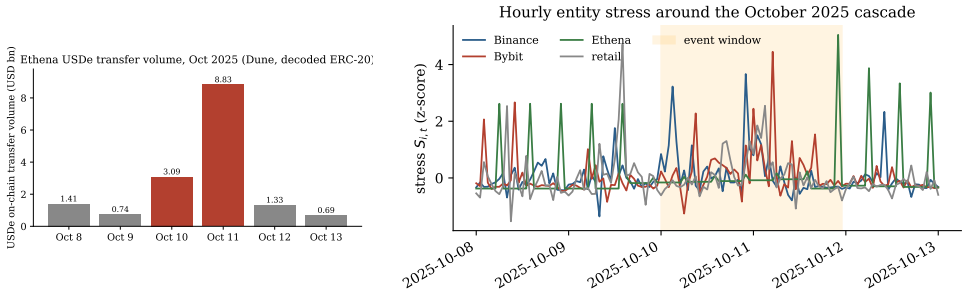



**Fig. 1** Left: on-chain USDe transfer volume around the October 2025 cascade (decoded ERC-20 transfers via Dune). Right: hourly entity stress states $S_{i,t}$ (shaded: the 10–11 October event window).

# 4 Data and Entity Resolution

## 4.1 Data, entity universe and stress states

The analysis uses public data only. Flows come from SQL over Dune Analytics' decoded `erc20_ethereum.evt_Transfer` table joined to the `labels.cex_ethereum` labels, prices and protocol total-value-locked (TVL) series from DefiLlama, and contract metadata from Etherscan. The cascade is visible in the raw transfers: on-chain volume of the synthetic dollar USDe rises from \$0.74bn on 9 October to \$3.09bn on 10 October and \$8.83bn on 11 October (Figure 1, left). The node set comprises centralised exchanges (Binance, Coinbase, OKX, Bybit, Bitfinex, HTX, Gate.io, Crypto.com, Bitget, Bullish), custodians (Copper, Ceffu, FalconX), DeFi protocols (Aave, Ethena, Lido, Maker/Sky), and an aggregate retail node that collects all unresolved addresses. Stress is operationalised by entity type and expressed as rolling $z$-scores: CEX and custodian stress combines the net on-chain outflow ratio with a transfer-volume surge; stablecoin-issuer stress combines the peg deviation $|1-p_t|$ with net redemption outflow; lending and liquid-staking stress combine TVL and price drawdowns; and retail stress is the market (ETH) drawdown. The settlement network is dominated by exchange–retail volume (Coinbase receives \$43.5bn from retail and sends \$43.4bn back); the smaller direct exchange-to-exchange transfers are recovered, and the SCM propagates along them (Figure 1, right).

**Table 1** Hypotheses evaluated in this paper.

| ID | Statement | Status |
|---|---|---|
| H1 | The observed on-chain flow graph predicts stress propagation more accurately than a graph estimated from price returns. | Not established (Section 5.1) |
| H2 | Flow direction carries predictive information (A3): reversing edges degrades prediction in directional events and is near-neutral in custodial settlement. | Partial, exploratory (Section 5.3) |
| H3 | OC-CoVaR identifies a different set of systemically important entities than $\Delta$CoVaR applied to returns. | Partial, six entities (Section 5.2) |
| H4 | The counterfactual measure decomposes realised losses into interpretable upstream channels. | Supported, descriptive (Section 5.2) |
| H5 | On-chain composability and collateral edges add loss-attribution information beyond price correlation. | Not established (Section 5.3) |

### 4.2 Entity resolution

A clustering error directly corrupts $\mathrm{pa}(i)$, since a mistake fabricates or severs a causal edge. We therefore use four tiers, building on Meiklejohn et al. (2013) and Weber et al. (2019). Tier 0 is the deposit-sweep heuristic (at least $\tau = 0.8$ of an address's outflow goes to an exchange hot wallet). Tiers 1 and 2 are a gradient-boosted classifier and a truncated-SVD embedding. Tier 3, the primary path, matches both transfer endpoints to curated seed labels and Dune's `labels.cex_ethereum`, collapsing sub-wallets to their exchange root. Each edge inherits the minimum confidence of its endpoints. The mint/burn sentinel is dropped first because its volume would dominate every volume-weighted statistic. Tiers 1–2 are validated in Appendix B, and their address-count metrics overstate economic recall, so Tier 3 remains the resolver.

## 5 Empirical Evaluation

Table 1 lists the hypotheses. Unless noted, results are at hourly resolution (the consolidated run: 600 fitting hours, 48 event hours, 17 entities, 39 directed edges; Figure 2). The daily specification is reported where it differs; backtests of the return-based inputs are in Appendix A. It is the only specification whose coefficients are archived, so the analytic checks in Section 5.5 refer to it.

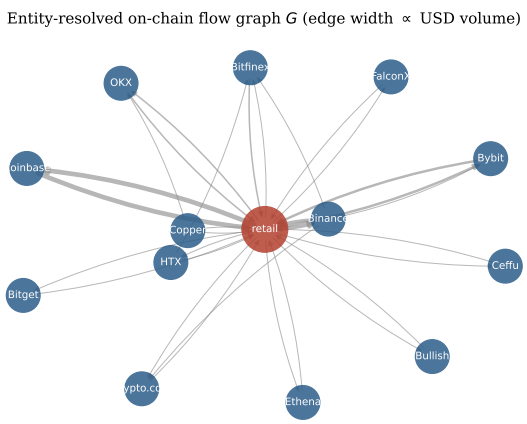



**Fig. 2** The entity-resolved on-chain flow graph $G$ recovered by the Tier-3 layer. Edge width is proportional to cumulative USD flow.

H2 was first framed as two claims, one for custodial and one for directional events. The event- and edge-type-conditional reading of H2 emerged from the analysis and should be read as exploratory. H5 was not tested directly: the evidence bearing on it is the daily Ethena→Aave channel and the edge-type split in Section 5.3.

## 5.1 Observed graph versus estimated graph (H1)

The graph is built from pre-event flows only. At daily resolution the price-estimated Granger graph predicts better (RMSE 1.35 against 1.92 over 42 predictions): with seven daily observations per entity, the denser estimated graph wins. At hourly resolution the ordering reverses. The observed graph attains RMSE 1.2529, against 1.2568 for the Granger graph and 1.2547 for the partial-correlation graph, a margin of about 0.1%. A paired bootstrap over the 288 predictions (2,000 i.i.d. resamples of the prediction-level weighted errors, which ignores serial dependence and so likely understates uncertainty) gives an observed-minus-Granger advantage in mean weighted error of 0.003 with 95% interval $[-0.012, +0.016]$, which includes zero, and the observed graph is better in 71% of resamples. With strictly lagged flows the comparison is essentially unchanged (1.258 against 1.247). On a leakage-free target built from price,

**Table 2** H1 head-to-head: event-window prediction error (RMSE; lower is better), identical VEIN machinery, varying only the graph source.

| | **Daily** ($n = 42$) | | **Hourly** ($n = 288$) | |
|---|---|---|---|---|
| **Graph source** | RMSE | corr | RMSE | corr |
| Observed on-chain | 1.92 | −0.05 | **1.2529** | 0.292 |
| Estimated (Granger) | **1.35** | 0.56 | 1.2568 | 0.255 |
| Estimated (partial-corr) | 1.65 | 0.48 | 1.2547 | 0.285 |

peg and TVL drawdowns, the Granger graph is better (1.36 against 1.68), which is expected for a target matched to a price-derived graph. We therefore treat H1 as suggestive, and the paper's main contribution does not rest on it.

## 5.2 Systemic ranking and counterfactual attribution (H3–H5)

The hourly OC-CoVaR exported-risk ranking (Table 3, Figure 3) is led by retail, Binance, Coinbase and Bitfinex. Over the six entities with a corresponding price series (retail–ETH, Binance–BNB, Ethena–ENA, Aave–AAVE, Lido–stETH, MakerSky–DAI), the Spearman correlation between the hourly exported-risk ranking and the ΔCoVaR ranking from hourly returns is 0.52 (Table 3). With six points this is an imprecise estimate and no significance is claimed. It indicates that the two rankings differ, as H3 requires, but not by how much. The daily specification gives 0.72 against ΔCoVaR and 0.52 against MES (`results/validation.json`), so the figure depends on the specification and the benchmark.

The daily ranking differs, and this matters for interpretation. At daily resolution Lido has the largest exported risk (4.06), although the forensic narratives document it as an absorber. The transmitter-versus-absorber check therefore fails at daily resolution (separation −1.97, pairwise concordance 0.50) and passes at hourly resolution (separation 12.87, concordance 1.00). External validity belongs to the hourly specification, and daily rankings should not be used to name transmitters.

**Table 3** Top six OC-CoVaR exported-risk entities (hourly) and ΔCoVaR (hourly returns) for the six mapped assets. Spearman $\rho = 0.52$ between the two rankings over the six mapped entities.

| Entity | Exported risk | Asset | ΔCoVaR |
|---|---|---|---|
| retail | 57.36 | ETH | −0.00400 |
| Binance | 24.52 | AAVE | −0.00395 |
| Coinbase | 17.35 | BNB | −0.00326 |
| Bitfinex | 11.42 | stETH | −0.00259 |
| Gate.io | 7.92 | ENA | −0.00258 |
| Bybit | 4.88 | DAI | +0.00078 |

The counterfactual decomposition (Table 4) recovers inter-exchange settlement channels. At hourly resolution Binance→Bybit accounts for 98.1% of Bybit's window distress, followed by Binance→Gate.io (58.4%) and Binance→Bitfinex (45.3%). Under Gaussian relative perturbation of every coefficient (standard deviation 15%, 40 draws), the Binance→Bybit share is $0.98 \pm 0.01$ and stays positive in all draws. At daily resolution, where composability edges dominate, the leading channel is Ethena→Aave (11.3%).

For Bybit we also compute an order-robust Shapley decomposition of the hinge game over Binance, Coinbase and Bitfinex (Figure 3, right), by exhaustive enumeration of orderings. In this game a candidate set is held at its observed stress and every other entity at its pre-crisis level, so only direct channels into Bybit operate. The values are Binance 3.41, Coinbase 0.00 and Bitfinex −0.14, which sum to the total effect 3.28 to within $5 \times 10^{-16}$. The negative Bitfinex value is not the intermediary credit of Proposition 4, which requires propagation through unselected entities. The hourly coefficients are not archived, so we do not trace its source further.

## 5.3 Edge-reversal falsification and the validity battery (H2)

Table 5 reports one-step weighted RMSE for the three graph variants in four settings: daily (seven event days, 119 entity-day predictions), the consolidated hourly run (816

**Table 4** Leading counterfactual loss-attribution channels $\Delta_i^{CF}(j)$.

| Resolution | Channel $i \to j$ | share | Interpretation |
|---|---|---|---|
| hourly | Binance → Bybit | 98.1% | inter-exchange settlement |
| hourly | Binance → Gate.io | 58.4% | inter-exchange settlement |
| hourly | Binance → Bitfinex | 45.3% | inter-exchange settlement |
| daily | Ethena → Aave | 11.3% | USDe collateral composability |
| daily | Binance → Aave | 4.2% | CEX-to-DeFi spillover |

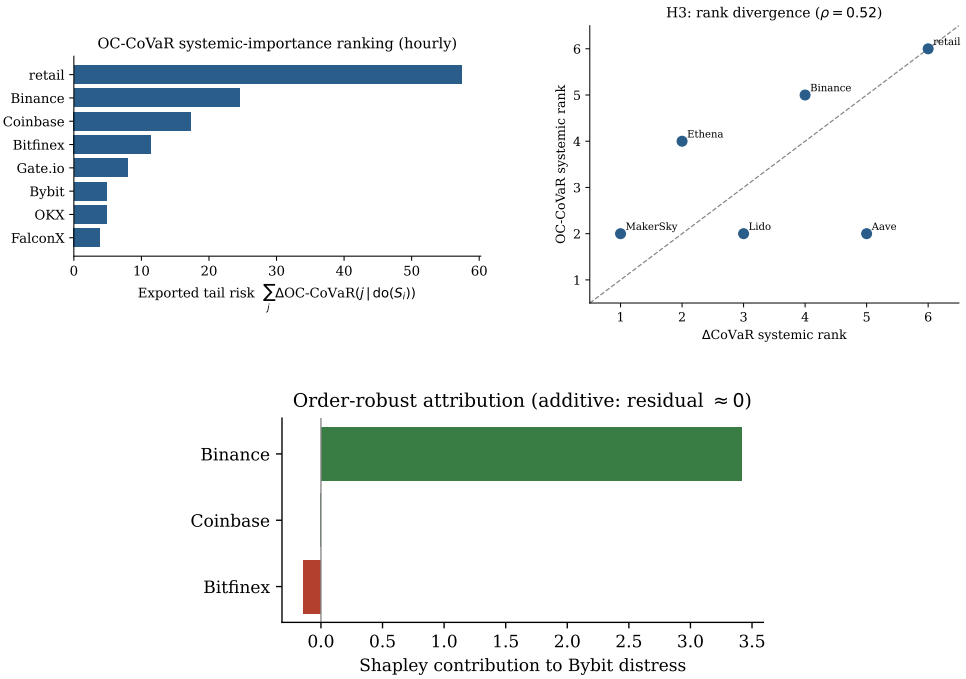



**Fig. 3** Top left: OC-CoVaR exported-risk ranking (hourly). Top right: rank divergence between OC-CoVaR and ΔCoVaR over the six mapped entities (hourly; Spearman $\rho = 0.52$). Bottom: Shapley decomposition of Bybit's event-window distress over Binance, Coinbase and Bitfinex (candidates at observed stress, all others at pre-crisis level); the values sum to the total effect (additive residual $\approx 10^{-16}$).

predictions), a second hourly run with 216 fitting hours (816 predictions), and a non-flow stress target at hourly resolution. The symmetric graph is included because it shows whether direction carries information beyond connectivity.

**Table 5** Edge-reversal falsification (one-step weighted RMSE; lower is better). The best graph in each column is bold.

| Graph | Daily | Hourly (consol.) | Hourly (216 fit h) | Non-flow target |
|---|---|---|---|---|
| true direction | 2.19 | **1.663** | 1.801 | 1.025 |
| reversed edges | **1.92** | 1.675 | **1.790** | 1.129 |
| symmetric | 1.94 | 1.667 | 1.797 | **0.942** |

The result is mixed. At daily resolution the test has little power and the reversed graph has the lowest error. In the consolidated hourly run the three graphs lie within 1.663–1.675, a near-tie. In the 216-fit run they lie within 1.790–1.801, and reversal is again marginally best. On the non-flow target, reversal raises the error by about 10% relative to the true graph (1.025 against 1.129), so direction matters relative to reversal. However, the symmetric graph is best on that target (0.942), so direction does not beat connectivity in every comparison. Splitting the consolidated run by edge type gives the same picture: reversing the 35 custodial edges raises the error to 1.675, while reversing the four composability edges leaves it at 1.663. The directional signal sits on custodial edges; the four composability edges are too low-volume to register in one-step prediction.

The validity battery addresses the remaining threats. The placebo check is an implementation check rather than evidence: unconnected pairs have zero attribution, as Corollary 2 requires. The confounding analysis gives moderate robustness values. The strongest edge, retail→Coinbase, has $RV = 0.226$, and Coinbase→retail has 0.175. Dropping the lowest-$RV$ edges leaves the ranking almost unchanged (Spearman 0.94, top three identical). A latent AR(1) shock common to all entities gives Spearman 0.84 at moderate and 0.81 at strong strength, with the top three unchanged. At hourly resolution, Tier-0 and Tier-3 graphs give Spearman 0.81 over six common entities ($p = 0.05$). The ranking places documented transmitters above documented absorbers, as the forensic post-mortems do (Ali 2025; Lim 2026), and this check holds at hourly resolution only.

Two further points bound the interpretation. The retail node absorbs every counterparty the heuristics fail to attach to a named entity, so it is "everything not yet resolved". Removing it leaves Binance as the top transmitter (16.0), followed by FalconX (2.4) and Ethena (2.1). Under Gaussian relative perturbation of every coefficient (standard deviation 15%, 25 draws), a deterministic zero-shock proxy of the exported-risk ranking has mean Spearman 0.88 with its unperturbed version (minimum 0.66, top-three overlap 0.79). The proxy's base ranking (retail, HTX, Bitfinex, Copper, FalconX) is not the Monte-Carlo ranking of Table 3, so this checks the stability of the method and not of the reported ranking.

## 5.4 Multi-event validation

We rerun the pipeline on two earlier episodes: the FTX collapse (event window 8–11 November 2022) and the USDC depeg around the Silicon Valley Bank failure (event window 10–13 March 2023), using the dollar assets present then and the same CEX label set (Appendix A). Table 6 reports the results. Transmitter rankings are sensible in each event without retuning: Binance and Coinbase lead in FTX, and the major exchanges lead in the SVB depeg. The reversal results are mixed. In FTX the true-direction graph is best. In SVB it beats the reversed graph but the symmetric graph is marginally best. In the October 2025 cascade the three graphs lie within 1.663–1.675. Direction is informative in FTX, where contagion spreads outward from a failing venue, and it is not decisive in SVB or in the custodial October cascade. These comparisons are low-powered: the earlier episodes have 7 and 9 pre-event fitting days and 44 and 48 one-step predictions, respectively.

## 5.5 Analytic checks on the daily specification

The numbers come from `results/analytics.json` (daily SCM, $T = 7$, 17 entities; spectral radius $\rho(A) = 0.276$). $B^+$ is $\Lambda_i(7)$ per unit $s^*$ (Corollary 3), so the stored

**Table 6** Multi-event validation: top OC-CoVaR transmitters and edge-reversal test across three stress episodes (one-step weighted RMSE; lower is better; best graph bold).

| Event | Top transmitters | true | reversed | symmetric | verdict |
|---|---|---|---|---|---|
| FTX (Nov 2022) | Binance, Coinbase, Crypto.com | **1.455** | 1.543 | 1.476 | true best |
| USDC/SVB (Mar 2023) | retail, Bybit, Binance, OKX | 1.567 | 1.610 | **1.564** | symmetric best |
| Oct 2025 cascade | retail, Binance, Coinbase | **1.663** | 1.675 | 1.667 | near-tie |

**Table 7** Flow-free bounds on exported risk, daily specification ($T = 7$). Rows with non-zero bound; the nine zero-bound entities are discussed in the text.

| Entity | Stored $\mathcal{E}$ (MC) | $B^+$ per unit $s^*$ | Implied min. $s^*$ | $\infty$-horizon multiplier |
|---|---|---|---|---|
| Lido | 4.06 | 1.37 | 2.97 | 0.228 |
| Bitget | 0.181 | 2.34 | 0.077 | 0.404 |
| retail | 0.111 | 0.428 | 0.259 | 0.0718 |
| Ethena | 0.0863 | 0.442 | 0.195 | 0.0733 |
| Bybit | 0.0696 | 0.705 | 0.0986 | 0.122 |
| Copper | 0.0430 | 1.46 | 0.0295 | 0.250 |
| Binance | 0.0325 | 0.0397 | 0.819 | 0.00754 |
| FalconX | 0.00767 | 0.0944 | 0.0813 | 0.0195 |

exported risk must satisfy $\mathcal{E}(i) \leq s^* B^+$, and the implied minimum $s^*$ is $\mathcal{E}/B^+$. The infinite-horizon multiplier (Corollary 3(ii)) is a reference quantity, not a bound.

There are no violations of the bound. Nine entities have $B^+ = 0$ and all have stored $\mathcal{E} = 0$. Only Aave and MakerSky lack outgoing edges in the archived graph. For the other seven, every archived outgoing coefficient rounds to zero at four decimals. This is a one-directional sanity check ($B^+ = 0$ implies $\mathcal{E} = 0$), not out-of-sample validation. Because coefficients are archived to four decimals, the implied minimum $s^*$ values are approximate. For Lido the stored exported risk is consistent with the bound only if $s^* \geq 2.97$; the hourly ranking gives Lido zero exported risk.

The certified ranking radius (Lemma 1) applies to the ranking by $\Lambda_i(T)$ under relative coefficient perturbations $\varepsilon$. For the top five entities (Bitget, Copper, Lido, Bybit, Ethena), the certified radii for consecutive pairs are 0.161, 0.027, 0.271 and 0.047, and no pair is certified at $\varepsilon = 1$. The certificate does not cover the exported-risk ranking $\mathcal{E}$, flows, shocks, $s^*$ or estimation error.

On a synthetic deviation path (seed 7), the closed-form walk decomposition agrees with subset enumeration to $5.6 \times 10^{-17}$. Subset enumeration needs $2^p$ oracle calls against $p!\,p$ for permutations: 8,192 against 80,951,270,400 at $p = 13$, the retail parent set. The exact subset computation for retail took 0.19 s on the machine used (timing is machine-dependent). The hourly coefficients are not archived, so none of these checks covers the hourly specification.

# 6 Discussion and Conclusion

Treating the observed transaction graph as a structural causal model yields a ranking that differs from ΔCoVaR (Spearman 0.52 over six entities, hourly) and a counterfactual decomposition of model-implied distress that is exact within the model. The graph predicts the October 2025 event competitively with the estimated-graph baseline, by a margin whose interval includes zero.

The directional assumption A3 is the most nuanced result. Reversing flow edges degrades prediction in FTX and on the non-flow target, but the symmetric graph is better than the true-direction graph in several comparisons. The evidence supports connectivity as the dominant signal and direction as an event-dependent refinement. VEIN should claim directed content where the mechanism is directional (collateral and composability links, outward propagation from a stressed hub) and connectivity-level content for symmetric settlement flows.

Four limitations bound these conclusions. CEX opacity is the main threat to A2: the proximate cause, an exchange's pricing engine, is off-chain, and the robustness values bound but cannot remove this threat. Resolution recall is the second: only about 1% of the $438bn of window flow moves directly between two named entities, and the volume-weighted recall of the Tier-1 classifier is 0.001 (Appendix B). Third, daily and hourly results disagree on the ranking, external validity holds only hourly, and hourly coefficients are not archived. Fourth, three episodes are a small basis for generality.

VEIN offers a ranking of which on-chain entities export stress and a counterfactual statement of what one entity's distress contributed to others' losses, both auditable from public ledger data. For supervisors and exchange or protocol risk teams, this offers a transparent complement to price-based monitoring: each attributed channel points to identifiable transfers that can be inspected, and the model-implied reading is stated explicitly. Next steps are intraday analysis across several events and an instrumented treatment of the CEX engine.

**Statements and Declarations. Data availability.** The repository [repository link withheld for review] contains all code, the SQL issued to Dune Analytics, and every derived result file (`results/*.json` and `results/*.md`) underlying the numbers, tables and figures. Raw Dune rows are not redistributed; the scripts regenerate them with a Dune API key. The underlying public data come from Dune Analytics (https://dune.com), DefiLlama (https://defillama.com) and Etherscan (https://etherscan.io), as described in Section 4. No restricted or proprietary data were used.

**Ethics approval and consent to participate.** Not applicable: this study uses only public, aggregated on-chain and market data and involves no human subjects.

**Consent for publication.** Not applicable.

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# Appendix A Implementation details and backtests

The pipeline is implemented in Python, with on-chain data queried from Dune Analytics and cached by query hash. Structural equations (2) are fitted by ridge regression with penalty $\lambda = 1$. The deposit-sweep threshold is $\tau = 0.8$, graph edges are kept above US$1M (daily) or US$0.1M (hourly) cumulative flow, the VaR level is $q = 0.95$,

and all Monte-Carlo and bootstrap procedures use seed 7. The H1 bootstrap resamples prediction-level errors i.i.d. The coefficient-perturbation checks multiply every stress and flow coefficient by $1 + \eta$ with $\eta \sim N(0, 0.15^2)$.

The consolidated hourly run fits the structural equations on 600 hours before 10 October 2025 and evaluates 48 event hours. It uses 150 Monte-Carlo paths per quantity (`run_best.py`), while the library defaults in `vein/risk.py` are 300–400. Severe-distress interventions set $s^*$ to each entity's 95th percentile of stress over the full sample window, which includes the event.

The reported runs resample residuals independently for each entity. The options `joint=True` (one row index shared by all entities) and `abduction="full"` (inverting the structural equations on the observed event path) are available but were not used for the reported numbers. The default `abduction="legacy_zero"` sets event-window shocks to zero, as residuals exist only on the pre-event days. The analytic checks use `run_analytics.py` and `tests/test_analytics.py`.

**Backtests.** A rolling historical VaR on daily returns ($n = 810$) has Kupiec (Kupiec 1995) $p = 0.24$ (BTC) and 0.58 (ETH), and Christoffersen (Christoffersen 1998) conditional-coverage $p = 0.11$ (BTC) and 0.001 (ETH). The hourly backtest ($n = 886$) has Kupiec $p = 0.03$ (BTC) and 0.11 (ETH).

## Appendix B Entity-resolution validation

Table B1 reports the held-out validation of the Tier-1 classifier against Dune's curated labels on the 7,841 event-window addresses, after the sentinel address is excluded. The base rate of CEX-labelled addresses is 0.5% (40 addresses). The held-out ROC-AUC is 0.98 and the average precision 0.24.

At its $F_1$-maximising threshold the classifier recovers a third of CEX-labelled addresses by count (precision 0.27, recall 0.33), but by USD volume recall is only 0.001.

**Table B1** Entity-resolution validation (Tier-1/2 against held-out Dune labels; mint/burn sentinel excluded).

| Metric | Value |
|---|---|
| Addresses (USDe event window) | 7 841 |
| CEX-labelled (base rate) | 40 (0.5%) |
| Tier-1 ROC-AUC (held out) | 0.98 |
| Tier-1 best-$F_1$ precision / recall (by address count) | 0.27 / 0.33 |
| Tier-1 precision / recall at the same threshold (by USD volume) | 0.21 / 0.001 |

It misses high-volume CEX addresses, which matter most for a dollar-denominated measure. Tier 3 therefore remains the production resolver.

# Appendix C Proofs

Let $h(y) = \max(y, 0)$, which is nondecreasing and 1-Lipschitz, so $\min(\delta, 0) \le h(x + \delta) - h(x) \le \max(\delta, 0)$.

**Proof of Proposition 2.** (a) Let $D_t = S_t^{(s)} - S_t^{(0)}$. Both paths share the initial state and the inputs $\kappa$, so $D_{-1} = 0$. For $j \neq i$, $D_{j,t} = \sum_p a_{jp} D_{p,t-1}$. For $i$, $D_{i,t} = s$ for $t < T$. We show $D_t = s\tau_t$ by induction. At $t = 0$, $D_0 = se_i = s\tau_0$. For $t \ge 1$, the $j \neq i$ coordinates satisfy $D_{j,t} = s \sum_p a_{jp} \tau_p(t-1) = s\,(A\tau_{t-1})_j = s\,\tau_j(t)$, since the reset affects only the $i$-th coordinate, and $D_{i,t} = s = s\,\tau_i(t)$.

(b) For a fixed realisation let $x_t = S_{j,t}^{(0)}$. Then $L_j^{(s)} - L_j^{(0)} = \sum_{t<T} [h(x_t + s\tau_{ji}(t)) - h(x_t)]$, and each summand lies in $[\min(s\tau_{ji}(t), 0), \max(s\tau_{ji}(t), 0)]$. The sums $\ell$ and $u$ of these endpoints are deterministic, so $L_j^{(0)} + \ell \le L_j^{(s)} \le L_j^{(0)} + u$ pointwise. Quantiles are monotone and translation-equivariant, which gives $\mathrm{VaR}_q(L_j^{(0)}) + \ell \le \mathrm{VaR}_q(L_j^{(s)}) \le \mathrm{VaR}_q(L_j^{(0)}) + u$, and the same holds for the Monte-Carlo estimator with shared draws. □

**Proof of Corollary 2.** The induction in the text shows that $\tau_{ji}(t) \neq 0$ requires a directed path from $i$ to $j$ of length at most $t$. Without such a path, $\tau_{ji}(t) = 0$ on the

window and Corollary 1 gives $\Delta\text{OC-CoVaR}_q(j \mid i) = 0$. For the attribution, $r_j \equiv 0$ in Proposition 3 with $K = \{i\}$, so (9) gives $\Delta_i^{CF}(j) = 0$. □

**Proof of Corollary 3.** (i) The upper bound in Proposition 2(b) is non-negative for $s > 0$, so $\max(\Delta\text{OC-CoVaR}_q(j|i), 0) \le s \sum_t \max(\tau_{ji}(t), 0)$. Summing over $j \neq i$ gives $\mathcal{E}(i) \le s\Lambda_i(T)$. So $\Lambda_i(T) = 0$ forces $\mathcal{E}(i) = 0$. (ii) For $t \ge 1$, $x_t = \tau_{\cdot i}(t)$ restricted to $V \setminus \{i\}$ satisfies $x_t = A'x_{t-1} + b$, with $A' = A_{-i,-i}$, $b = A_{-i,i}$, $x_0 = 0$. With $m = (I - A')^{-1}b$ and $\rho(A') < 1$, we have $x_t - m = A'^t(x_0 - m) \to 0$. The Cesàro mean of the convergent sequence $\max(x_t, 0)$ converges to $\max(m, 0)$, which gives the limit of $\Lambda_i(T)/T$. □

**Proof of Proposition 3.** Let $D = S^K - S^{\mathrm{f}}$. Both paths share the inputs $\kappa$ and the initial state, so $D_{-1} = 0$. For $k \in K$, $D_{k,t} = p_{k,t} - S^{\mathrm{f}}_{k,t} = -g_{k,t}$. For $j \notin K$, both paths satisfy (6), so $D_{j,t} = \sum_p a_{jp} D_{p,t-1}$. Induction on $t$ shows $D = -r$, since $r$ satisfies the same recursion with the same forcing. Hence $S^K = S^{\mathrm{f}} - r(K)$. Taking $K = \{i\}$ and $p_{i,t} \equiv S_i^{\mathrm{pre}}$ gives $L_j^{\mathrm{do}} = \sum_t \max(S^{\mathrm{f}}_{j,t} - r_{j,t}(\{i\}), 0)$, and (9) follows from $\Delta_i^{CF}(j) = L_j^{\mathrm{fact}} - L_j^{\mathrm{do}}$. □

**Proof of Proposition 4.** (a) Unrolling the recursion for $r$ backwards from $(j, t)$, a node in $K$ stops the recursion as a forcing term and a node outside $K$ passes it to its parents. Hence $r_{j,t}(K)$ is the sum of $\omega(w)$ over walks from a forced node $x_0 \in K$ whose intermediate nodes avoid $K$, which is the indicator in (a). Walks that revisit $x_0$ contain $x_0 \in K$ among their intermediates and are excluded by the same condition, so restricting to $\mathcal{W}$ changes nothing.

(b) By linearity it suffices to treat $u_w(K) = \mathbf{1}[x_0 \in K,\ I(w) \cap K = \emptyset]$. Let $k = x_0$, $I = I(w)$, $m = |I|$, and let $\mathrm{Pre}_y$ be the players preceding $y$ in a uniformly random ordering; the Shapley value of $y$ is its expected marginal contribution. For $y = k$ the marginal is $\mathbf{1}[\mathrm{Pre}_k \cap I = \emptyset]$, and $k$ precedes all of $I$ with probability $1/(m+1)$. For $y \in I$ the marginal is $-\mathbf{1}[k \in \mathrm{Pre}_y,\ \mathrm{Pre}_y \cap I = \emptyset]$. This requires $k$ to be first and $y$ second among $\{k\} \cup I$, which has probability $1/((m+1)m)$. For any other $y$ the game

does not depend on $y$. This gives $\phi_k(u_w) = 1/(1+m)$, $\phi_y(u_w) = -1/(m(m+1))$ for $y \in I$, and zero otherwise. Summing over walks with the weights $\omega(w)$ gives (10).

(c) Summing (10) over $k \in P$ gives $\omega(w)[\frac{1}{1+m} - \frac{1}{m+1}] = 0$ for $m \geq 1$ and $\omega(w)$ for $m = 0$, which matches (a) at $K = P$.

(d) If $S^{\mathrm{f}}_{j,t} \geq 0$ and $S^{\mathrm{f}}_{j,t} - r_{j,t}(K) \geq 0$, then both hinge terms are linear, so $v_H(K) = v(K)$. □

**Lemma 1** (Certified ranking radius) *Let $\tilde{A}$ satisfy $|\tilde{a}_{jp} - a_{jp}| \leq \varepsilon |a_{jp}|$ with the same sparsity pattern, and let $\tilde{\Lambda}_i$ be $\Lambda_i(T)$ computed from $\tilde{A}$. Then $|\tilde{\Lambda}_i - \Lambda_i(T)| \leq R_i(\varepsilon)$, where $R_i(\varepsilon) = \sum_{j \neq i} \sum_{t<T} \left[\tau_{ji}^{(1+\varepsilon)|A|}(t) - \tau_{ji}^{|A|}(t)\right]$. If $\Lambda_r - \Lambda_{r+1} > R_r(\varepsilon) + R_{r+1}(\varepsilon)$, the order of the pair is preserved under every such perturbation.*

*Proof* $\tau_{ji}(t)$ is a sum over walks from $i$ to $j$ of length at most $t$ avoiding $i$ in between. The telescoping bound $|\prod \tilde{a}_e - \prod a_e| \leq \prod(1+\varepsilon)|a_e| - \prod |a_e|$ gives the entrywise bound, and $h$ is 1-Lipschitz. □

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